Run AI on someone else’s computer without them seeing your input

Stays on your computer	Goes to each server
Your actual text	A random-looking piece of it
The final answer	The public model + random masks

One server you control is enough

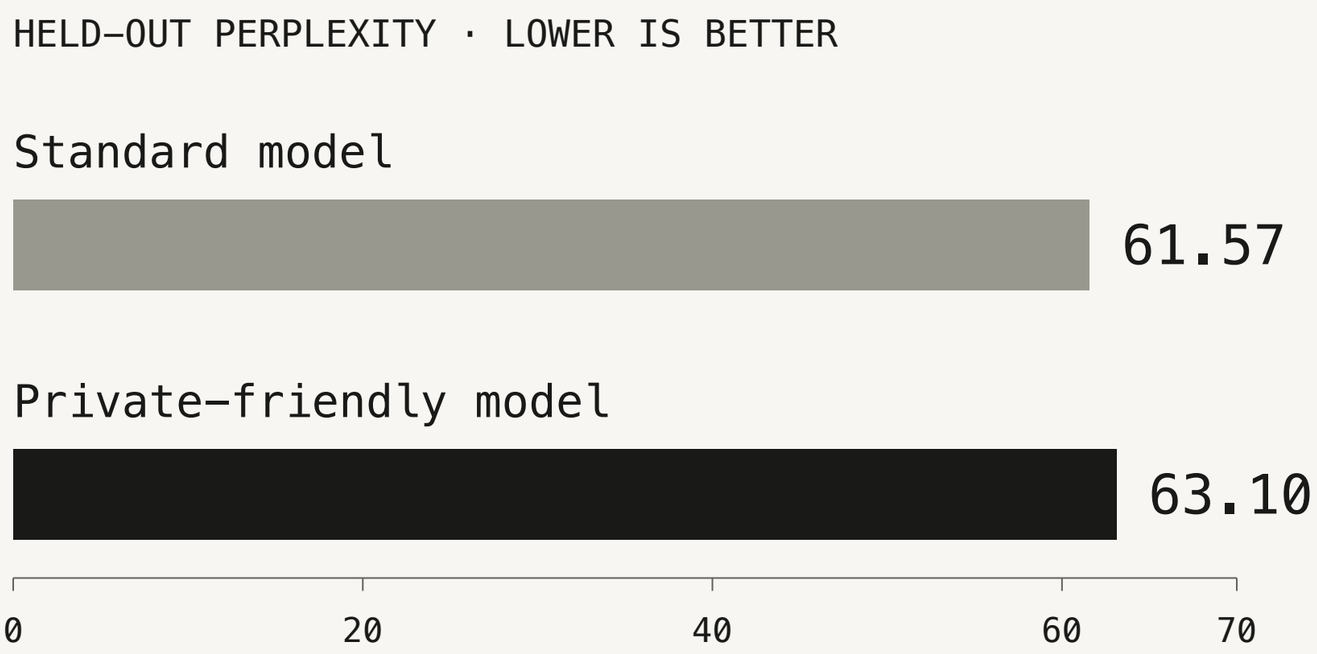
If these servers’ pieces are combined	Can your input be read?
Any one server	No
Any two servers	No
All three servers	Yes
One of the three is yours	No

If all three are combined in secret, you can’t detect it. So run one yourself.

Assumes the servers are run by different people

In the demo, all three run on our desk

In a separate test, the private-friendly model was 2.48% worse



6,164,096
parameters in each test model

14,336,000
training tokens

Not the demo model.
About quality, not speed.

Measured: perplexity on held-out text · data/quality.json

Separate from the demo

The model only adds, multiplies and squares

Step	Who does it	Messages
Split the input into pieces	Your computer	Pieces go to the servers
Multiply by public weights	Each server, on its own	None needed
Square	Servers, through your computer	Checked before it’s used
Multiply again, then check	Servers, then your computer	Answer pieces come back
Per character		2 round trips

From engine/device.py and engine/node.py

Privacy costs extra compute and messages

What works now, and what’s next

Works now	Next
A 19,295-parameter note model	Real-size language models
Three servers, over the network or USB	Servers run by different people
Same output as the normal model	A full security review
Your computer makes the random masks	Servers make them and talk directly

Run one server yourself. Use other people’s for the rest.